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(54) METHODS FOR FORMING MULTI-LAYERED IMAGE JIGSAW PUZZLES AND RELATED DEVICES

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(57) **ABSTRACT**

Multi-layered image jigsaw puzzles include images on opposing surfaces of an optically clear and transparent structure to allow each image to be clearly viewed from either surface. The images on both surfaces of the puzzle form a complimentary image.

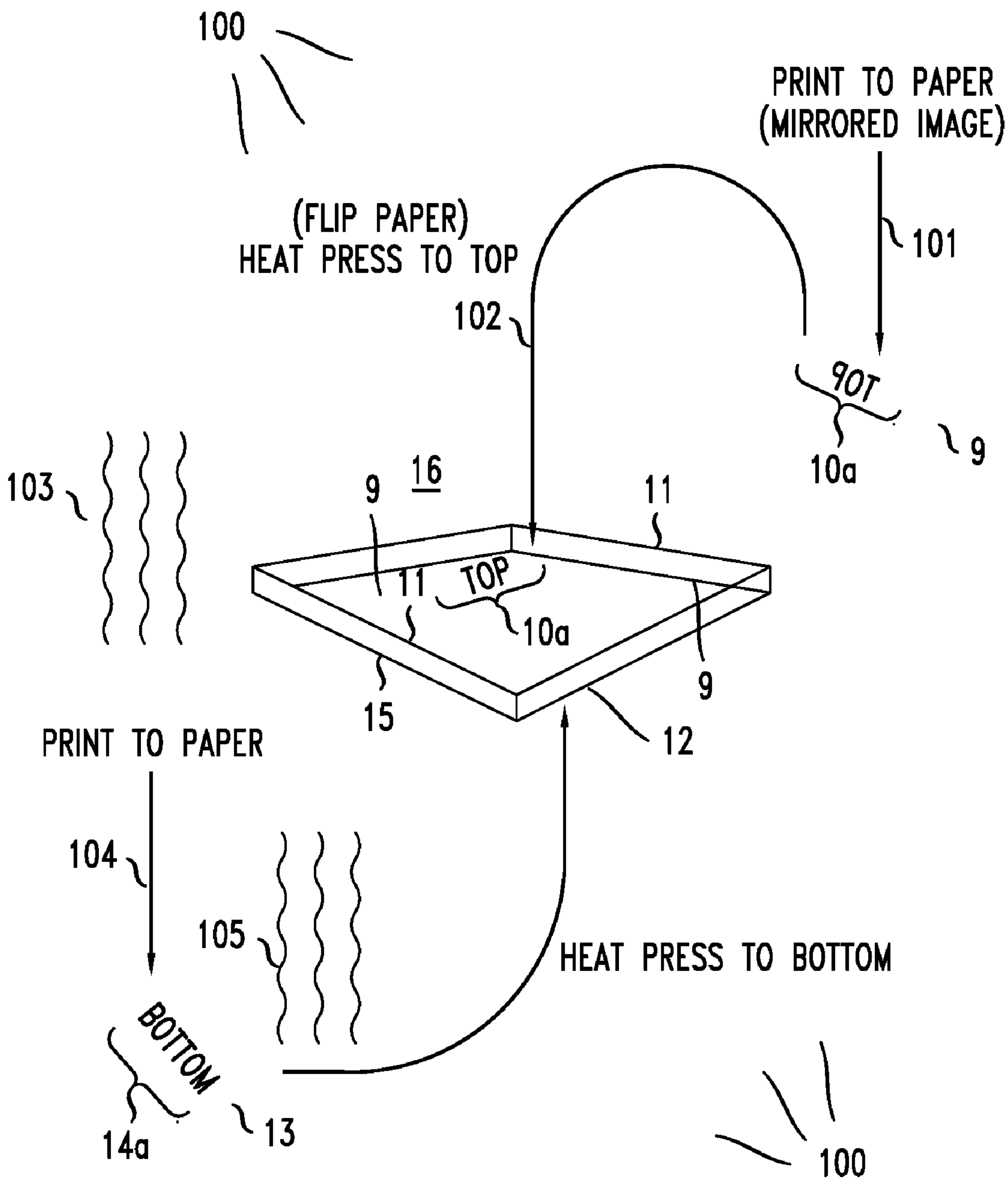


FIG. 1A

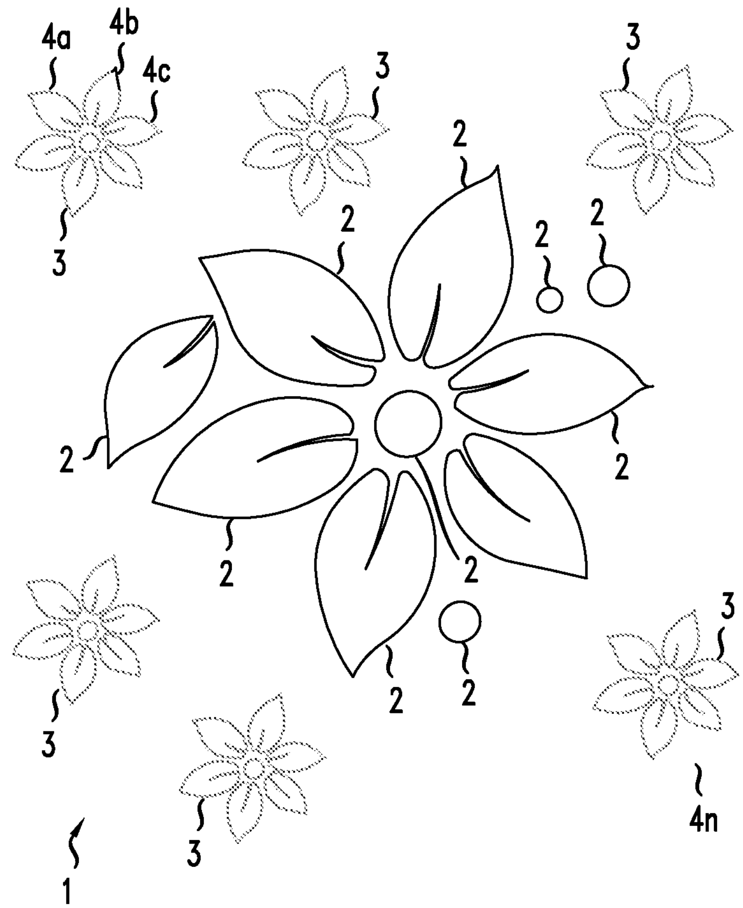


FIG. 1B

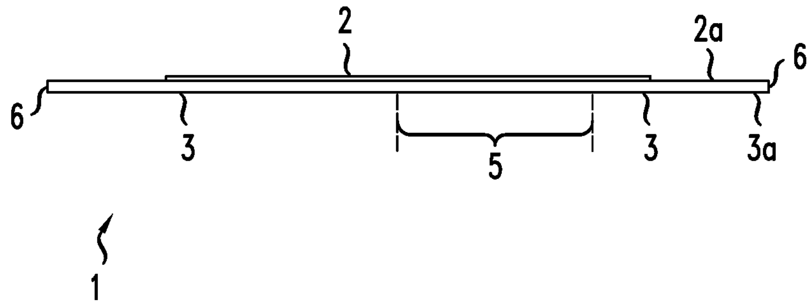


FIG. 2

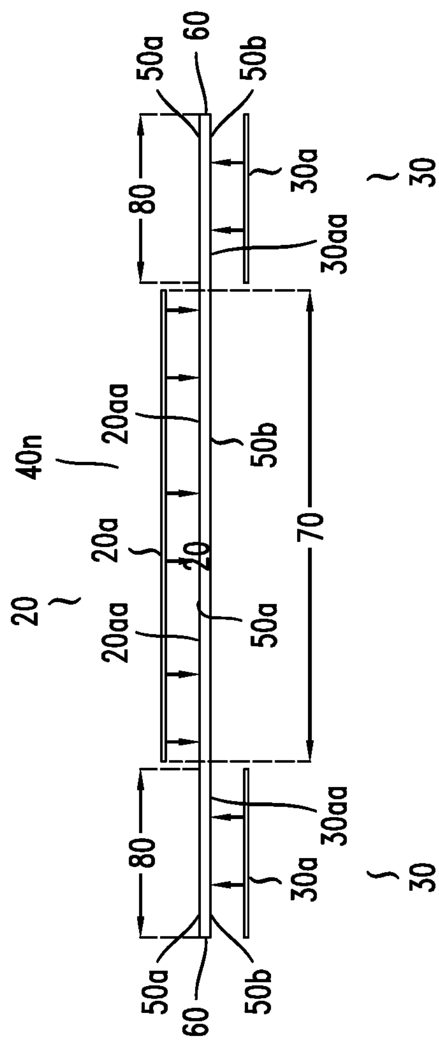

50a
60
50b
80
30a
30
30aa
20aa
40n
50b
20a
20
70
20
20aa
50a
30aa
80
30a
30
50a
60
50b

FIG. 3A

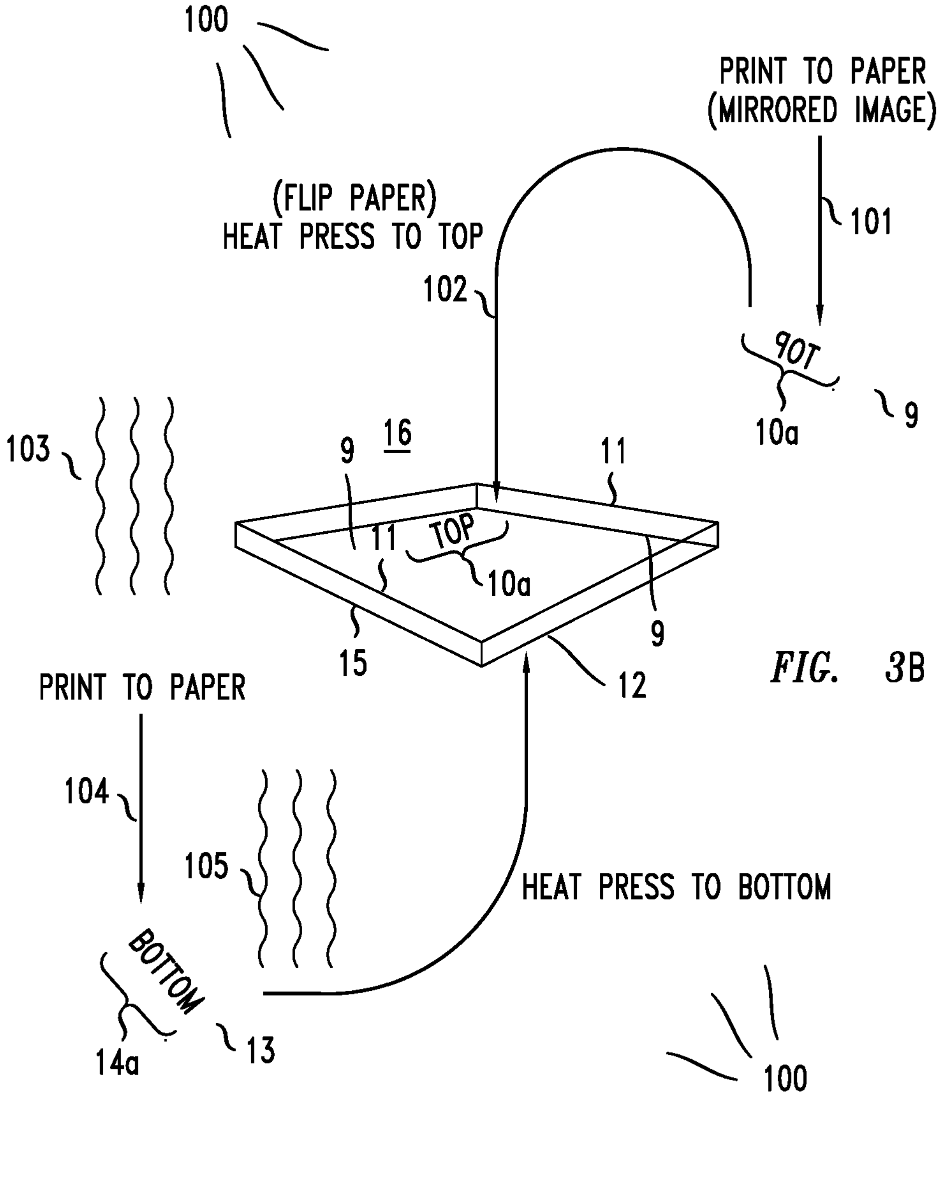


FIG. 3B

FIG. 3C

FIG. 4A

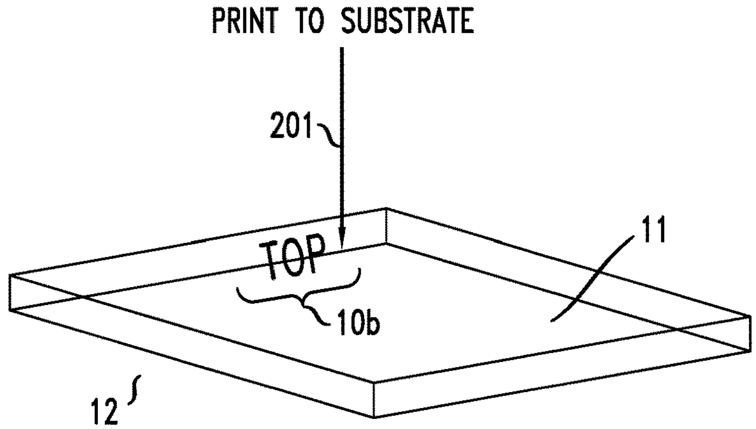


FIG. 4B

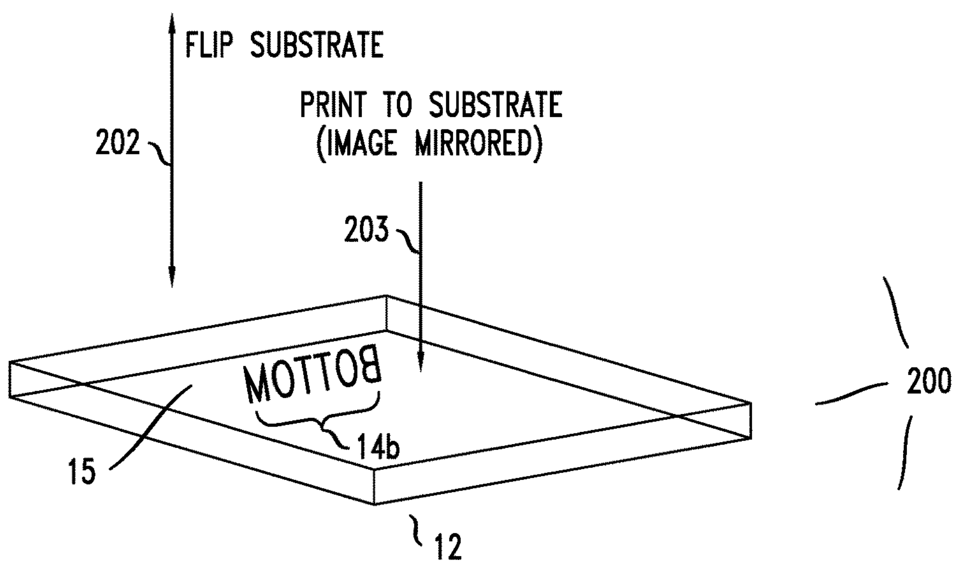


FIG. 4C

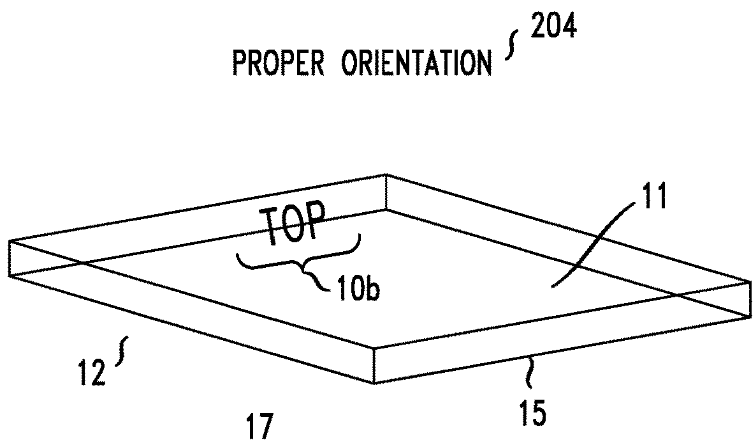
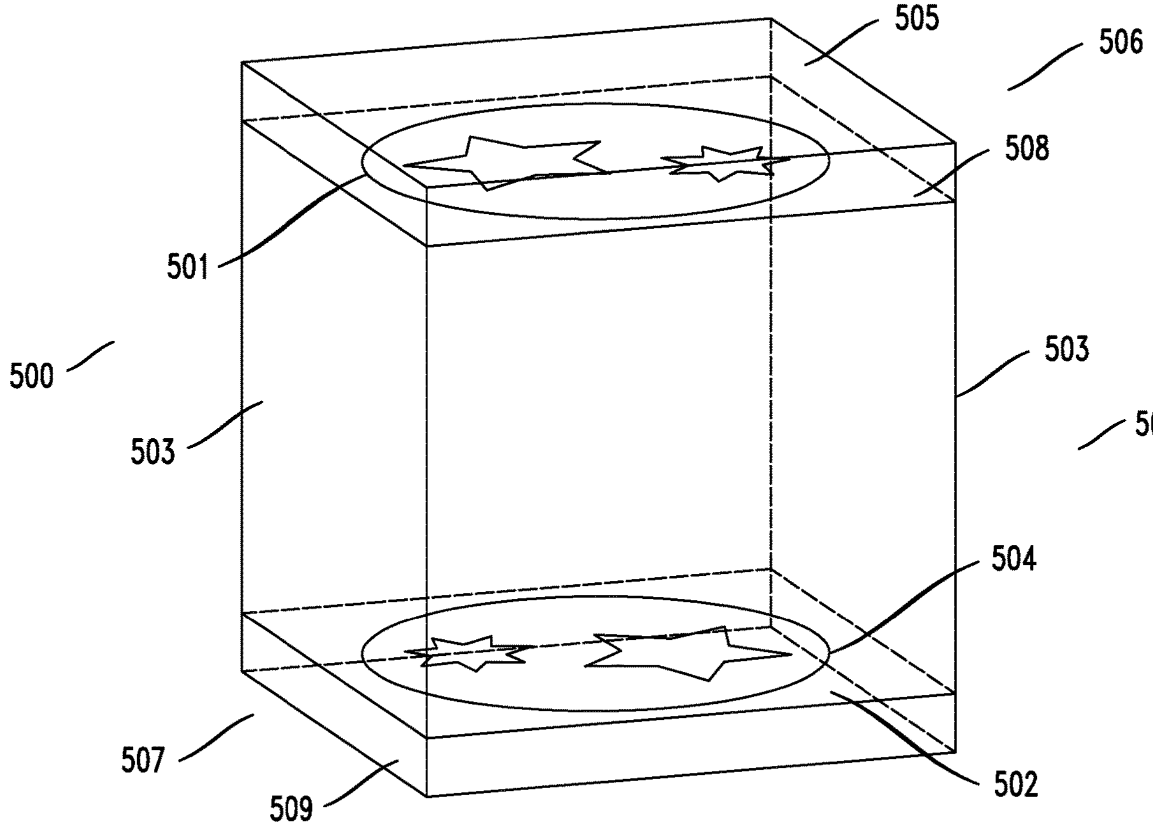


FIG. 5



METHODS FOR FORMING MULTI-LAYERED IMAGE JIGSAW PUZZLES AND RELATED DEVICES

RELATED APPLICATION

[0001] This application claims priority to U.S. Provisional Application 63/562,747 filed Mar. 8, 2024 (the "'747 Application"). This application incorporates by reference the entire disclosure of the '747 Application as if it were set forth in full herein.

INTRODUCTION

[0002] This section introduces aspects that may be helpful to facilitate a better understanding of the present disclosure. Accordingly, the statements in this section are to be read in this light and are not to be understood as admissions about what is, or what is not, in the prior art.

[0003] The present application relates to methods for forming multi-layered image jigsaw puzzles and other devices with images on at least the front and back of the puzzle or device, where the images on the front and back can be seen from both the front and back on and/or through a transparent substrate, and form a layered, or background and foreground effect.

[0004] To date, existing methods have attempted to form jigsaw puzzles and other devices with images on at least the front and back. However, both images cannot be seen from either side and, therefore, do not form a multi-layer effect, e.g. background and foreground image effect.

[0005] Still other puzzles are made of a structure composed of a clear material. However, either no image is placed on both sides of the clear material or if an image is placed, it is only placed on one side of the material, not both and, therefore, does not render a multi-layered image puzzle.

[0006] Accordingly, there is a need for better methods for forming multi-layered jigsaw puzzles and other devices with images on at least the front and back that overcome the disadvantages of existing methods.

SUMMARY

[0007] The inventor has discovered methods and related devices for forming multi-layered puzzles (e.g., jigsaw puzzles). For example, exemplary inventive methods for forming multi-layered image puzzles (again, for example, jigsaw puzzles) may comprise forming a first image on a first surface of a clear, transparent structure or substrate (collectively "structure") and forming a second image on a second surface of the clear, transparent structure, wherein the first and second images may be separated from one another and form a complimentary image and at least the first and second surfaces form multiple layers. Similarly, inventive multi-layered image jigsaw puzzles may comprise multiple, stacked, transparent structures with images on surfaces of the structures, and each structure is separated from another structure such that the images form a complimentary image with at least two sides or planes of a structure forming multiple layers.

[0008] The inventor has discovered a number of illustrative devices that include inventive puzzles.

[0009] For example, one device is a puzzle, such as a jigsaw puzzle, that may comprise: a first surface of an optically clear, transparent structure, and a second surface of the optically clear, transparent structure, wherein the first surface comprises a first image disposed thereon and the second surface comprises a second image disposed thereon, and where the first and second images form a clear, complimentary image viewed through both surfaces and at least the first and second surfaces form multiple layers.

[0010] It should be understood that such a puzzle may comprise a plurality of interlockable pieces, wherein one or more of the interlockable pieces of the plurality of interlockable pieces may comprise one or more image segments of the first image on a first surface of a respective piece and one or more image segments of the second image on a second and opposing surface of the respective piece.

[0011] Alternatively, such a puzzle may comprise one or more of the interlockable pieces of the plurality of interlockable pieces comprises one or more image segments of the first image on a first surface of a respective piece or one or more image segments of the second image on a second and opposing surface of the respective piece.

[0012] Still further, in another alternative embodiment such a puzzle may comprise one or more of the interlockable pieces of the plurality of interlockable pieces comprises no image segments of the first image on a first surface of a respective piece and no image segments of the second image on a second surface of the respective piece.

[0013] In embodiments, in each of the puzzles described above (and elsewhere herein) such a second surface may be parallel to, and opposite, the first surface.

[0014] In many embodiments described above and elsewhere herein the first and second surfaces may be separated from one another.

[0015] In addition to the puzzles described above the present inventor discloses a number of related methods. One such method comprises a method for forming a puzzle, such as a jigsaw puzzle. Such a method may further comprise forming a first image or a first image segment on a first surface of an optically clear, transparent structure (e.g., an acrylic structure); and forming a second image or a second image segment on a second surface of the clear, transparent structure, wherein the first and second images or their respective segments form a complimentary image and at least the first and second surfaces form multiple layers.

[0016] In an embodiment, such a method may further comprise applying a mirrored version of the first image or first image segment onto a first material; applying the first material with the applied mirrored version of the first image or first image segment onto the first surface of the optically clear and transparent structure such that the first image or first image segment is non-mirrored; applying heat to the first material containing the first image or first image segment or to the first surface of the structure to create a bond between the first material and the structure; applying the second image or the second image segment onto a second material; applying the second material with the applied second image or second image segment onto the second surface of the optically clear and transparent structure; and applying heat to the second material containing the second image or second image segment or to the second surface of the structure to create a second bond between the second material and the structure.

[0017] In an embodiment, the first material and the second material may comprise a specially-coated, heat transfer paper.

[0018] Still further, such a method may comprise: applying the first image or first image segment directly on the first

surface of the structure by directing ultra-violet responsive ink representing the first image or first image segment onto the first surface of structure; and applying a mirrored version of the second image or second image segment directly on the second surface of the structure **12** by directing ultra-violet responsive ink representing the second image or second image segment onto the second surface of the structure.

[0019] The inventor also discloses a number of devices that include inventive jigsaw puzzles. One such device may comprise a cube-shaped device comprising: a first surface of a first substrate composed of a first optically clear, transparent structure, comprising a first image disposed thereon; and a second, opposing surface of a second substrate composed of a second optically clear, transparent structure, comprising a second image disposed thereon, and where the first and second images form a clear, complimentary image, wherein the first and second surfaces and each of the images are separated from one another and form opposing sides of the cube-shaped device. It should be understood that the first and second structures may comprise puzzles, such as jigsaw puzzles.

[0020] The inventor believes that the inventive methods and related devices described herein will allow for the formation of multi-layered jigsaw puzzles and other devices with images on at least the front and back.

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] FIGS. **1**A and **1**B depict different views of an illustrative, exemplary inventive multi-layered image puzzle in accordance with an embodiment of the present disclosure.

[0022] FIG. **2** depicts a simplified illustration of elements of an image segment of an exemplary inventive multi-layered image puzzle in accordance with an embodiment of the present disclosure.

[0023] FIGS. **3**A to **3**C depict respective diagrams of an exemplary method for forming inventive multi-layered image jigsaw puzzles in accordance with an embodiment of the present disclosure.

[0024] FIGS. **4**A to **4**C depict respective diagrams of a second exemplary method for forming inventive multi-layered image jigsaw puzzles in accordance with an embodiment of the present disclosure.

[0025] FIG. **5** depicts an illustration of a cube-shaped device that includes one or more jigsaw puzzles according to an embodiment of the disclosure.

DETAILED DESCRIPTION, WITH EXAMPLES

[0026] Exemplary embodiments for forming inventive multi-layered image puzzles are described herein and are shown by way of example in the drawings. Throughout the following description and drawings, like reference numbers/characters refer to like elements. It should be understood that although specific embodiments are discussed herein, the scope of the disclosure is not limited to such embodiments. On the contrary, it should be understood that the embodiments discussed herein are for illustrative purposes, and that modified and alternative embodiments that otherwise fall within the scope of the disclosure are contemplated.

[0027] It should be noted that one or more exemplary embodiments may be described as a process or method (the word "method" may be used interchangeably with the word "process" herein). Although a process/method may be described as sequential, it should be understood that such a process/method may be performed in parallel, concurrently or simultaneously. In addition, the order of each step within a process/method may be re-arranged. A process/method may be terminated when completed, and may also include additional steps not included in a description of the process/method if, for example, such steps are known by those skilled in the art.

[0028] It should be understood that when a system or device (e.g., a structure), or a component or element of a system or device, is referred to, or shown in a figure, as being "connected" to (or other tenses of connected) another system, device (or component or element of a system or device) such systems, devices, components or elements may be directly connected, or may use intervening components or elements to aid a connection. In the latter case, if the intervening systems, devices, components or elements are well known to those in the art they may not be described herein or shown in the accompanying figures for the sake of clarity.

[0029] It should be understood that, as used herein, the designations "first", "second", "third", "fourth" and "top" and "bottom" are used to distinguish one parameter or element from another and does not indicate a priority or order. The parameters or elements could be re-designated (i.e., re-numbered) and it would not affect the methods or devices described herein.

[0030] As used herein the phrases "top" and "bottom" are relative terms used to distinguish a view of elements shown in a figure and are not intended to limit the orientation or direction of those elements. Consequently, depending on how a structure is held or viewed, the top and bottom of a structure can be perceived as reversed.

[0031] As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items. As used herein, the singular forms "a," "an" and "the" are intended to include the plural form, unless the context and/or common sense indicates otherwise. It should be understood that the terms "comprises", "comprising,", "includes" and/or "including", when used herein, specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or combinations thereof. As used herein the term "configured to" means "functions to" unless the context, common sense or knowledge of one skilled in the art indicates otherwise.

[0032] As used herein the phrases "embodiment" or "exemplary" mean a non-limiting example of the present disclosure. Though the embodiments described herein and shown in the figures may depict certain geometric shapes, it should be understood that these shapes are merely exemplary, and, accordingly, other shapes may be substituted for the so described and depicted shapes.

[0033] As used herein the word "clear", "clearly" or its tenses means visually transparent e.g., the opposite of opaque.

[0034] As used herein the phrase "jigsaw puzzle" means a structure that comprises uniquely shaped, interlockable elements (i.e., pieces) that when separated, challenge a person to assemble the elements into a cohesive image of interlocked elements, and where usually, but not always, the elements may be colored.

[0035] As used herein the phrase "multi-layered image jigsaw puzzle" means a jigsaw puzzle that includes two or more (a "plurality") of images, where each image is on a different, substantially parallel layer or plane within a transparent structure. As used herein the phrase "multi-layered" means multiple layers.

[0036] Referring now to FIGS. **1**A and **1**B there are illustrated top and side views of an exemplary inventive multi-layered image jigsaw puzzle **1** in accordance with an embodiment of the present disclosure.

[0037] As shown in FIG. **1**A the inventive puzzle **1** may comprise a plurality of interlockable jigsaw puzzle elements or pieces **4***a* to **4***n* (where "n" represents the last jigsaw puzzle piece). When each of the pieces **4***a* to **4***n* is interlocked with at least one other piece, the interlocked jigsaw pieces **4***a* to **4***n* form the puzzle **1**.

[0038] In the embodiments shown in FIGS. **1**A and **1**B the puzzle **1** may comprise a first surface **2***a* of an optically clear, transparent structure **6** having a first image **2** (e.g., an artistically designed image) applied onto its surface and interlockable pieces **4***a* to **4***n* (e.g., top surface) and a second surface **3***a* of the optically clear, transparent structure **6** having a second image **3** (e.g., an artistically designed image) applied onto its surface and interlockable pieces **4***a* to **4***n* (e.g., bottom surfaces).

[0039] In an embodiment, when the pieces are connected to the structure **6** the second surface is typically parallel to, and opposite, the first surface. In an embodiment, each of the surface on opposite sides of the structure **6** form a different layer of the puzzle **1**. Having images on two surfaces, the puzzle **1** may be referred to as a multi-layered image jigsaw puzzle. Further, each of the surfaces of the structure, and therefore each of the images **2**,**3** may be separated (distance-wise) from one another due to, for example, a thickness of the structure. Alternatively an air gap or another gap (not shown in figure) may be created within the structure in between the two opposing surfaces. Yet further, if needed, miniature mechanical supports may be inserted into the structure between the two surfaces to support a given surface or surfaces.

[0040] In an embodiment, both images **2**,**3** may be visible from either side of the puzzle **1**. Said another way, the puzzle **1** may be configured to allow both images **2**,**3** to be clearly viewed through both surfaces and from either side of the puzzle (e.g., when the puzzle is laying horizontally, both images can be seen) due to the fact that the structure **6** may be composed of an optically clear and transparent material upon which the images may be attached to or formed on a respective surface.

[0041] In an embodiment, when viewed from one side the image **2** on one surface may be in the foreground while image **3** on another surface may be in the background (see FIG. **1**B). Alternatively, when viewed form an opposite side the image **2** may be in the background while the image **3** may be in the foreground. Further, in an embodiment, both images may be selected to form an overall image when the images are combined. Said another way, both images visually complement one another when combined on the same material (i.e., a "complimentary image").

[0042] In sum, an inventive jigsaw puzzle **1** may comprise a first surface of an optically clear, transparent structure **6**, and a second surface of the optically clear, transparent structure **6**, wherein the first surface comprises a first image **2** disposed thereon and the second surface comprises a second image **3** disposed thereon, and where the first and second images form a clear, complimentary image viewed through both surfaces and at least the first and second surfaces form multiple layers.

[0043] In an embodiment, one or more of the interlockable jigsaw puzzle pieces **4***a* to **4***n* having first surfaces that form the first image **2** may, or may not, be the same as the one or more jigsaw puzzle pieces **4***a* to **4***n* having second surfaces that form the second image **3**. Because some of the pieces **4***a* to **4***n* may have two different images on opposite surfaces, for the sake of clarity we will refer to the pieces that form both images as **4***a* to **4***n*.

[0044] In an embodiment, the first image **2** may comprise one or more first image segments formed on a first surface (e.g., "top" surface) of an optically clear and transparent structure **6** and the second image **3** may also comprise one or more second image segments formed on a second surface (e.g., "bottom" surface) of the structure **6**. In an embodiment, the structure **6** may be composed of an optically clear and transparent acrylic material, for example. The thickness of the acrylic material may vary depending on a desired application. In two non-limiting embodiments, the thickness may be ⅛ of an inch, or $1/16^{th}$ of an inch, for example. In an embodiment, the images **2**, **3** may be clearly seen on the optically clear and transparent structure **6** when viewed from either side. In an embodiment, each of the surfaces forms a different layer of the structure **6**.

[0045] Referring now to FIG. **2** there is depicted a single jigsaw puzzle piece **40***n* (where "n" indicates one jigsaw puzzle piece) which may be part of the puzzle **1**. As illustrated, piece **40***n* may comprise one or more first image segments **20***a* of a first image **20** formed on a first surface **50***a* (e.g., "top" surface) of an optically clear and transparent structure **60** and one or more second image segments **30***a* of a second mage **30** formed on a second surface **50***b* (e.g., "bottom" surface) of the structure **60**. In an embodiment, the structure **60** may be composed of an optically clear and transparent acrylic material, for example. The thickness of the acrylic material may vary depending on a desired application. In two non-limiting embodiments, the thickness may be ⅛ of an inch, or $1/16^{th}$ of an inch, for example. In an embodiment, the opposing image segments **20***a*, **30***a* (as well as opposing non-image segments discussed below) on surfaces **50***a*,**50***b* may be separated (distance-wise) from one another and may be clearly seen on the optically clear and transparent structure **60** when viewed from either side. In an embodiment, each of the surfaces **50***a*, **50***b* forms a different layer of the structure **60**.

[0046] In addition, in an embodiment, upon being formed on the structure **60** the image segments **20***a* and **30***a* form a combined, complimentary image.

[0047] Though FIG. **2** depicts a single first image segment **20***a* it should be understood that this is merely exemplary and that a given puzzle piece (such as pieces **40***n* or **4***a* to **4***n*) may comprise more than one first image segment on its first surface or none at all. Similarly, though FIG. **2** depicts two second image segments **30***a* it should be understood that this is merely exemplary and that a given puzzle piece may comprise a single second image segment, more than two second image segments or none at all on its second surface.

[0048] That is to say, in various embodiments, (i) one or more interlockable pieces of the plurality of interlockable pieces **4***a* to **4***n* may comprise one or more image segments of the first image **20** on a first surface **50***a* of a respective piece **4***a* to **4***n* (e.g., **40***n*) and one or more image segments of the second image **30** on a second and opposing surface

50*b* of the respective piece, or (ii) one or more of the interlockable pieces of the plurality of interlockable pieces **4***a* to **4***n* comprises one or more image segments of the first image **20** on a first surface **50***a* of a respective piece or one or more image segments of the second image **30** on a second and opposing surface **50***b* of the respective piece, or (iii) one or more of the interlockable pieces of the plurality of interlockable pieces **4***a* to **4***n* comprises no image segments of the first image **20** on a first surface **50***a* of a respective piece and no image segments of the second image **30** on a second surface **50***b* of the respective piece.

[0049] Those portions of a surface that include an image segment may be referred to as "image portions" while those portions that do not include at least one image segment may be referred to as "non-image portions". In FIG. **2**, the first surface **50***a* includes two non-image portions **80** while the second surface **50***b* includes one non-image portion **70**. For the sake of clarity, the portion of surface **50***a* upon which image segment **20***a* is applied can be referred to as an image portion **20***aa* and the portions upon which image segments **30***a* are applied can also be referred to as image portions **30***aa*.

[0050] In FIG. **2**, the non-image portion **70** is illustrated between image portions **30***aa*. Accordingly, in an embodiment each of the plurality of image portions **30***aa* may be separated by a non-image portion **70** (i.e., though this is optional). Said another way, a given puzzle piece **40***n* may comprise a portion (or portions) of a first or second surface **50***a*, **50***b* that does not have an image segment formed thereon (see also portion **5** on the bottom of puzzle **1** in FIG. **1**B). It should be noted that an exemplary puzzle may not include pieces that include a non-image portion. Said another way, all of the pieces of an exemplary puzzle may comprise image portions without any non-image portions.

[0051] In an embodiment, though a surface may include a non-image portion, such as portion **70** on second surface **50***b*, an opposing image segment and its respective opposing image portion on an opposing surface (i.e., on a different layer), such as segment **20***a* and portion **20***aa* on a first surface **50***a*, may be clearly visible and viewed through portion **70** because the structure **60** is optically clear and transparent.

[0052] Referring now to FIGS. **3**A to **3**C and **4***a* to **4**C there are illustrated two methods of forming a jigsaw puzzle where both methods include the steps of (i) forming a first image or a first image segment on a first surface of an optically clear, transparent structure; and (ii) forming a second image or a second image segment on a second surface of the clear, transparent structure, wherein the first and second images or their respective segments form a complimentary image and at least the first and second surfaces form multiple layers.

[0053] Specifically, FIGS. **3**A to **3**C illustrate a simplified diagram of an exemplary method **100** for forming an inventive multi-layered image jigsaw puzzle **16**. It should be understood that the method **100** that follows may be applied to a complete image or to an image segment.

[0054] Referring to FIG. **3**A, during step **101** a mirrored version of a first image segment or first image **10***a* may be applied to (printed onto) a first material **9**, by a light source (laser printer). In an embodiment, the mirrored version of the first image segment or image **10***a* may first be generated by a computer system (not shown) connected to the light source (e.g., laser printer; not shown) before being applied to the first material **9** by mirroring the orientation of an original image (e.g., "TOP").

[0055] In more detail, the first material **9** may comprise a specially-coated, heat transfer paper that may be applied to a substantially smooth and hard surface.

[0056] Thereafter, during step **102** the first material **9** comprising the imprinted, mirrored version of the first image segment or image **10***a* may be applied to a first surface **11** (e.g., top surface) of an optically clear and transparent acrylic structure **12** such that the first image segment or image **10***a* of the original image (e.g., TOP) can be correctly viewed and seen (i.e., a non-mirrored image) (see FIG. **3**B). Next, during step **103** energy (e.g., thermal energy, heat) may be applied to the first material **9** containing the first image segment or first image **10***a* and/or the first surface **11** of the acrylic structure **12** in order to create an adhesive bond between the first material **9** (upon which the first image segment or image **10***a* has been formed) and the acrylic structure **12**.

[0057] Continuing, during step **104** a second image segment or image **14***a* (e.g., "BOTTOM") may be applied to (printed onto) a second material **13** by a light source (e.g., laser printer, not shown) (see FIG. **3**C). In an embodiment, the second image segment or image **14***a* may first be generated by a computer system (not shown) connected to the light source before it is applied to the second material **13**.

[0058] In an embodiment, the second material **13** may also comprise a specially-coated, heat transfer paper that may be used on a substantially smooth and hard surface.

[0059] During step **105** the second material **13** comprising the imprinted version of the second image segment or image **14***a* may be applied to the second surface **15** (e.g., bottom surface) of the optically clear and transparent acrylic structure **12** such that the second image segment or image **14***a* (e.g., BOTTOM) can be viewed and seen (see FIG. **3**B).

[0060] During the same step or another step energy (e.g., thermal energy, heat) may be applied to the second material **13**, image segment or image **14***a* and the second surface **15** of acrylic structure **12** in order to create an adhesive bond between the second material **13** (upon which the second image segment or image **14***a* has been applied) and the acrylic structure **12**.

[0061] Backtracking somewhat, in more detail, during steps **103** and **105** as energy (heat) is being applied to either the first or second materials **9**, **13** the substance (e.g., ink) that makes up the first and second image segments or images **10***a*, **14***a* may be separated from a respective material **9**,**13** and transferred onto acrylic structure **12**.

[0062] Similar to before, the opposing images **10***a*, **14***a* on surfaces **11**,**15** may be separated (distance-wise) from one another.

[0063] In embodiments, the thickness of the structure **12** may again be ⅛ of an inch, or $1/16^{th}$ of an inch, for example, and both image segments or images **10***a*, **14***a* may be clearly seen on the optically clear and transparent structure **12**. In an embodiment, upon applying materials **9**,**13** to surfaces **11**,**15** the resulting structure forms a multi-layered image structure **12**.

[0064] In an embodiment, upon being applied to the structure **12** the image segments **10***a* and **14***a* may form a combined, complimentary image.

[0065] Alternatively, instead of applying (e.g., imprinting) the first and second image segments or images on a paper, the images may be directly formed on an acrylic structure itself through a method **200** that makes use of ultraviolet (UV) printing means (e.g. UV light). In embodiments, the wavelength of a UV light source that may be used by an inventive method may depend on the ink used and type of light source used (e.g., laser diode versus LED). In embodiments, the wavelength of a laser diode or LED light source configured to cure or dry such ink instantly as it is applied to a prepared substrate may range between 200 nanometers (nm) and 405 nm for laser diode light sources and 385 nm and 395 nm for LED light sources, with a preferred wavelength being 350 nm for a laser diode light source to name just one of many preferred wavelengths.

[0066] Referring now to FIGS. **4**A to **4**C there is illustrated a simplified diagram of an alternative, exemplary method **200** for forming an inventive multi-layered image jigsaw puzzle **17** using UV light. It should be understood that the method **200** that follows may be applied to a complete image or to an image segment.

[0067] In more detail, during step **201** (see FIG. **4**A) a first image segment or image **10***b* may be directly applied to a first surface **11** (top surface) of an optically clear and transparent acrylic structure **12** by, for example, a UV light source (not shown) that directs UV responsive ink representing the first image or first image segment **10***b* onto the first surface **11** of the structure **12**. Further, during step **201** the UV light source may apply UV light to the first surface **11** and ink to simultaneously and instantly cure the UV responsive ink to the structure **12**.

[0068] Thereafter, during step **202** the structure **12** may be flipped or turned so that a second surface **15** is now facing the UV light source (see FIG. **4**B). Alternatively, during step **202** the structure **12** may remain as oriented during step **201** but the UV light source is moved so that it is now able to direct UV light towards the second surface **15**.

[0069] In either case, during step **203** (see FIG. **4**B) a mirrored, second image segment or image **14***b* may be directly applied to the second surface **15** (bottom surface) of the same optically clear and transparent acrylic structure **12** by, for example, the UV light source (not shown) that directs UV responsive ink that represents the second image or second image segment **14***b* onto the structure **12**. Further, during step **203** the UV light source may apply UV light to the second surface **15** and ink to simultaneously and instantly cure the UV responsive ink to the structure **12**.

[0070] Thereafter, during step **204** the structure **12** may then be turned or otherwise oriented such that the first surface **11** is now the top surface (see FIG. **4**C) so that both first and second image segments or images **10***b*, **14***b* now form a combined, complimentary image when viewed from the top.

[0071] In an embodiment, upon applying images **10***b*,**14***b* to surfaces **11**,**15** the resulting structure forms a multi-layered image structure **12**.

[0072] Similar to before, the opposing images **10***b*,**14***b* on surfaces **11**,**15** may be separated (distance-wise) from one another, Further, in embodiments, the thickness of the structure **12** may again be ⅛ of an inch, or 1/16th of an inch, for example, and both image segments or images **10***b*, **14***b* may be clearly seen on the optically clear and transparent structure **12**.

[0073] When FIGS. **3**A to **3**C and **4**A to **4**C represent images instead of image segments, after methods **100** or **200** are completed, the acrylic structure (i.e., substrate) **12** may be cut or otherwise separated into the plurality of jigsaw puzzle pieces (e.g., pieces **4***a* to **4***n*). In an embodiment, the cutting or separation may be completed by a CO_2 laser cutting tool in order to produce puzzle pieces that fit precisely together and interlock.

[0074] In more detail, the following method may be used to cut the structure **12**.

[0075] In an embodiment, a low adhesion, minimum or no residue masking tape may be applied to both sides (i.e., first and second or top and bottom) of the structure **12**. The application of the masking tape helps protect the surfaces of both sides and their respective images from debris or deposits that are created when small pieces of the structure are removed and scattered as the structure **12** is cut into the jigsaw puzzle pieces (e.g., pieces **4***a* to **4***n*).

[0076] Once the masking tape is applied, the structure **12** may be inserted or otherwise positioned on a cutting machine, such as a CO_2 laser cutter (e.g., manufactured by Glowforge Pro). Thereafter, a user may input specific settings into the cutting machine to cut the structure **12** that comprises images on each of its sides into a plurality of jigsaw puzzle pieces (e.g., **4***a* to **4***n*).

[0077] Yet further, upon completing the cutting process the masking tape is removed from each jigsaw puzzle piece.

[0078] It should be understood that the inventive puzzles **1**, **16** and **17** in FIGS. **1**A to **4**C which include the features described herein may form a number of different shaped puzzles. That is, though shown as rectangular the inventive puzzles may be formed in a number of shapes, such as triangular, circular and hexagon to name just a few examples.

[0079] In addition, the first and second opposing surfaces of an inventive puzzle may be separated by more than ⅛ of an inch, or 1/16th of an inch. For example, such surfaces may be separated by a distance such that the opposing surfaces form surfaces of a cube (side-to-side).

[0080] Referring now to FIG. **5** there is depicted a device **500** that includes one or more inventive jigsaw puzzles **506**, **507**. In this embodiment the first surface **505** of an optically clear, transparent substrate **508** of the first puzzle **506** may comprise a first image **501** applied to its surface while the second, opposing surface **502** of a second optically clear, transparent substrate **509** of the second puzzle **507** may comprise a second image **504** applied to its surface, where both images can be viewed when looking through either surface (through the top or bottom surface, or side-to side). Other shaped puzzles may also be formed by separating the first and second surfaces.

[0081] In slightly more detail, the first and second substrates **508**, **509** may be composed of an acrylic for example.

[0082] In an embodiment, the combination of the first and second images **501**,**504** form a clear, complimentary image viewed through both surfaces and at least the first and second surfaces **502**,**505** form multiple layers.

[0083] In addition, in an embodiment the first and second surfaces **502**, **505** and therefore each of the images **501**, **504** may be separated (distance-wise) from one another due to, for example, a distance corresponding to the length of another side **503** of the cube-shaped device **500**, for example.

[0084] It should be noted that each surface may be considered a side of the cube-shaped device **500**, such that the opposing surfaces **502**,**505**, their corresponding images **501**, **504** and substrates **508**,**509** form opposing sides of the cube-shaped device **500**.

[0085] Each of the puzzles **506**,**507** described above may be formed by one or more of the methods described elsewhere herein, for example.

[0086] The description above has focused on exemplary embodiments for creating inventive multi-layered image puzzles. While certain embodiments have been described it should be understood that the inventive methods and devices are not limited to the described embodiments. Other variations of the inventive methods and devices may be implemented.

[0087] It should be understood that the features described above with respect to one embodiment illustrated by one figure, for example, may also be incorporated into another embodiment described herein and/or shown in a different figure. For the sake of clarity, however, such features have not been repeated for each other embodiment.

[0088] Numerous changes and modifications to the embodiments disclosed herein may be made without departing from the general spirit of the disclosure, the scope of which is best defined by the claims that follow.

I claim:

1. A puzzle comprising:

a first surface of an optically clear, transparent structure, and

a second surface of the optically clear, transparent structure, wherein the first surface comprises a first image disposed thereon, and the second surface comprises a second image disposed thereon, and where the first and second images form a clear, complimentary image viewed through both surfaces and at least the first and second surfaces form multiple layers.

2. The puzzle as in claim **1** wherein the puzzle comprises a plurality of interlockable pieces.

3. The puzzle as in claim **2** wherein one or more of the interlockable pieces of the plurality of interlockable pieces comprises one or more image segments of the first image on a first surface of a respective piece and one or more image segments of the second image on a second and opposing surface of the respective piece.

4. The puzzle as in claim **2** wherein one or more of the interlockable pieces of the plurality of interlockable pieces comprises one or more image segments of the first image on a first surface of a respective piece or one or more image segments of the second image on a second and opposing surface of the respective piece.

5. The puzzle as in claim **2** wherein one or more of the interlockable pieces of the plurality of interlockable pieces comprises no image segments of the first image on a first surface of a respective piece and no image segments of the second image on a second surface of the respective piece.

6. The puzzle as in claim **1** wherein the second surface is parallel to, and opposite, the first surface.

7. The puzzle as in claim **1** wherein the first and second surfaces are separated from one another.

8. The puzzle as in claim **1** wherein the puzzle comprises a jigsaw puzzle.

9. A method for forming a puzzle comprising:

forming a first image or a first image segment on a first surface of an optically clear, transparent structure; and

forming a second image or a second image segment on a second surface of the clear, transparent structure, wherein the first and second images or their respective segments form a complimentary image and at least the first and second surfaces form multiple layers.

10. The method as in claim **9** further comprising:

applying a mirrored version of the first image or first image segment onto a first material;

applying the first material with the applied mirrored version of the first image or first image segment onto the first surface of the optically clear and transparent structure such that the first image or first image segment is non-mirrored;

applying heat to the first material containing the first image or first image segment or to the first surface of the structure to create a bond between the first material and the structure;

applying the second image or the second image segment onto a second material;

applying the second material with the applied second image or second image segment onto the second surface of the optically clear and transparent structure; and

applying heat to the second material containing the second image or second image segment or to the second surface of the structure to create a second bond between the second material and the structure.

11. The method as in claim **10** wherein the first material and the second material comprises a specially-coated, heat transfer paper.

12. The method as in claim **9** further comprising:

applying the first image or first image segment directly on the first surface of the structure by directing ultra-violet responsive ink representing the first image or first image segment onto the first surface of structure; and

applying a mirrored version of the second image or second image segment directly on the second surface of the structure by directing ultra-violet responsive ink representing the second image or second image segment onto the second surface of the structure.

12. The method as in claim **9** wherein the structure is composed of an acrylic.

13. The method as in claim **9** wherein the puzzle comprises a jigsaw puzzle.

14. A cube-shaped device comprising:

a first surface of a first substrate composed of a first optically clear, transparent structure, comprising a first image disposed thereon; and

a second, opposing surface of a second substrate composed of a second optically clear, transparent structure, comprising a second image disposed thereon, and where the first and second images form a clear, complimentary image,

wherein the first and second surfaces and each of the images are separated from one another and form opposing sides of the cube-shaped device.

15. The cub-shaped wherein the first and second structures comprise jigsaw puzzles.

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